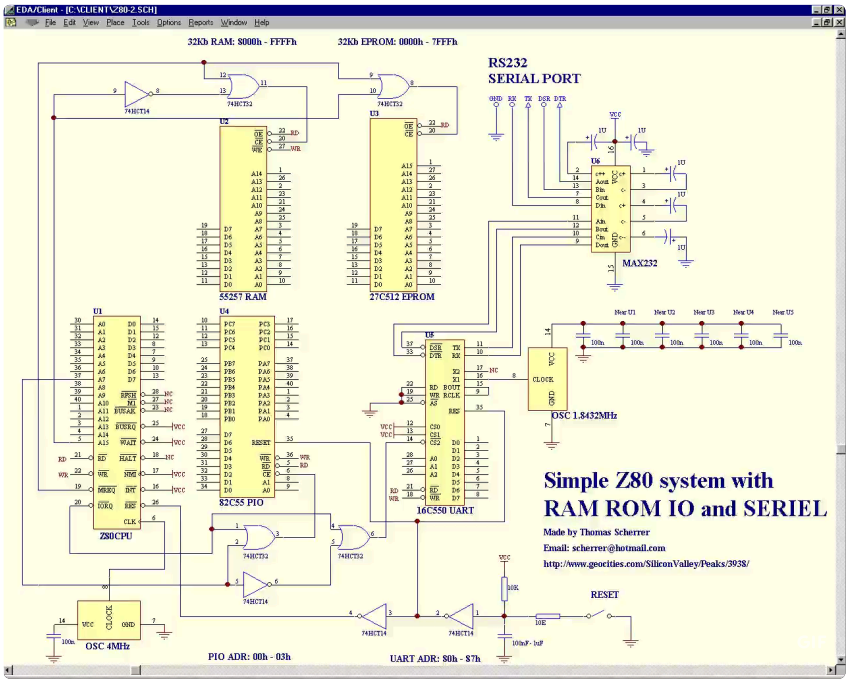
r/homebrewcomputer • 3y ago
u/lynchaj

Simple homebrew computer for beginners

Hi

I've been thinking what would be a good starter homebrew computer project for someone brand new to the hobby and wants to get started with something simple and a high probability of success.

In the past, I've recommended Z80.info as a good place to start, especially the circuit below which I think is almost foolproof and good for beginners. The circuit is old and could use a bit of updating. Also, it could use some additional debugging features to help new builders get started.



The design is quite limited in capability. It won't run CP/M or RomWBW. In many ways, it is more like an embedded microcontroller than a general purpose computer. It has a fixed 32KB ROM to boot, a fixed 32KB RAM for data and program storage. An Intel 8255 PPI for general purpose IO and a 16550 UART for serial (TTL and RS-232 levels). Multiple LEDs to capture useful signals and a connector for the PPI GPIO pins. Uses common 2.1mm 9VDC center-positive power supply with reverse voltage protection.

For software, it would run the Z80 debug monitor program which you could use to load other programs in Intel Hex format. It has many commands to explore and experiment with the system.

What I'd like to know is if I capture this circuit in KiCAD and make some PCBs, would anyone be interested in building these? Note: I am not offering to sell the boards or kits. Rather looking for a small group of people to build and initial design and send me their feedback. Again, not offering them for sale just for homebrew computer hobbyists entertainment purposes only.

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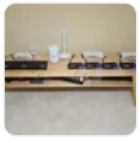
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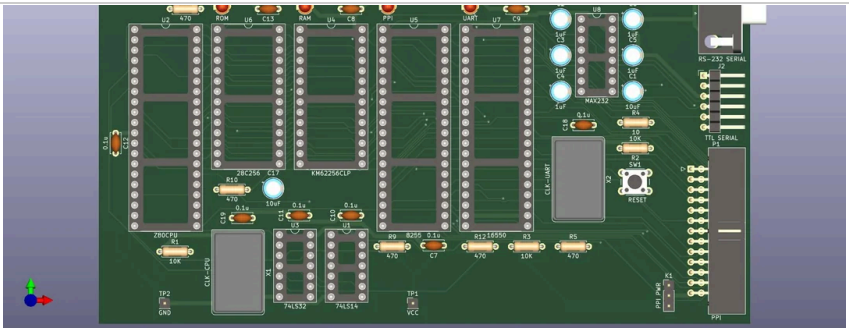


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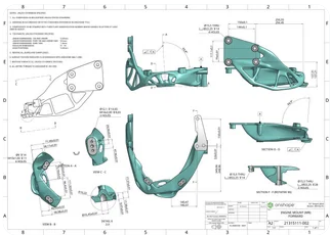


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rehsd • 3y ago

I'm always up for projects like this. 🙂 I'd be happy to build one for myself and provide feedback. Z80 has been on my list for a while.

4 Reply

lynchaj OP • 3y ago • Edited 3y ago

OK, I am going to spend a few days re-examining the design and making a new schematic in KiCAD and layout a PCB. I think the main goal should be ease of construction for new builders. There are several things that can be done.

Will update original post with some preliminary ideas.

Thanks!

2 Reply

Tom0204 • 3y ago

As an experienced Z80 guy, here are my tips:

Don't use the 8255 or the 16550. As far as I'm aware they're not mass-produced anymore so it would be better to go with the Zilog PIO and UART/SIO chips (though a PIO isn't really essential).

Swap the EPROM for EEPROM because if you plan on writing any BIOS or simple OS code, you'll be re-flashing the ROM a lot.

Apart from that it's still a pretty good design!

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r/HomeServer • 3y ago

Beginner looking for a simple linux setup

That's a very good description but it's also what a real computer should be. Machines like this are totally unencumbered by bloated operating systems and commercial software. They're a blank canvas that you paint by making your own software and hardware.

lynchaj OP • 3y ago

OK, thanks for the feedback. The updated design does use a 28C256 EEPROM for ease of programming. However, I am thinking to use a ZIF socket for the ROM because it is frequently changed during programming cycles and can easily be damaged. Also the EEPROM allows for in-circuit programming.

The target audience I have in mind for this board design are the homebrew computer beginners just to take the first move and get familiar with the design and build concepts. The FirstZ80 is not intended as a final destination but rather a first step. If beginners want to use other systems to get started, that is OK with me. More power to them whatever system they start with. Odds are, they will be successful regardless of their choice.

Unless you're referring to the original NMOS Intel 8255 PPI and 16550 UARTs, those are two of the most common LSI parts ever produced. If you include pin-compatible CMOS and licensed (and unlicensed) clones, I don't think we will ever run out of them or even have a shortage. They are available from nearly every electronic vendor and on the secondary market. I buy most of my parts from AliExpress and eBay but Mouser, Jameco, Digikey, etc. all carry these parts and have for many years.

That being said, I am a big fan of the Zilog Peripherals (PIO, SIOx, CTC, DMA, etc.) since they can communicate with the Z80 CPU via IM2. However, I would not recommend them for a first project since they can be tricky to program and finicky. They work great and I have a board dedicated to the Zilog Peripherals as part of my duodyne project. I recommend it as the default console.

Another bit of feedback regarded the use of interrupt. In retrospect, I think I can add a simple IM1 interrupt for the UART back to the CPU. Since there are only two peripherals (PPI and UART) and the PPI rarely involves an interrupt, I think I can dedicate the interrupt to servicing the UART. However, I've noticed that using interrupts with the 16550 is not efficient and most people would get better performance using the auto-handshaking modes with regular polling. Especially with a transmit/receive buffer like with a 16550 or better UART.

I'll store the design files here on the GitHub repository in case people want a closer look. Later today I will update the design regarding the ZIF socket and an IM1 interrupt.

https://github.com/lynchaj/duodyne/tree/main/!Development/FirstZ80

istarian • 3y ago • Edited 3y ago

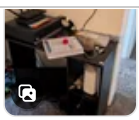
Texas Instruments still makes/ sells 16C550 chips?

https://www.ti.com/product/TL16C550C

You can buy 82C55 compatible parts on Mouser, albeit they're pretty expensive and most are PLCC packages.

https://www.mouser.com/c/?q=82C55

Beginner Homelabber here! This is my setup :)



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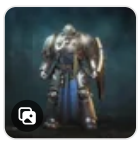
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Might as well use a CMOS Z80 in that case, though. Not sure what this design intended.

otr_trucker • 3y ago

I'd recommend John's Basement Z80 Retro Build

He has many videos explaining the layout of the board and what each chip does. He also explains writing the assembly code to test each feature of the board. And then he gives details instructions for porting CP/M.

production-dave • 3y ago

If you haven't already, have a look at Grant Searle's website

http://searle.wales/

He has a simple 7chip z80 design that runs basic. For something a little more complex you get cp/m.

production-dave • 3y ago

Also, I noticed the z80.info design does nothing with interrupts. They just tied high. For learning, I think it's important to allow an interrupt source even if its just a jumper to vcc when not used. It would be a shame to ignore them altogether.

SkyHighGhostMy • 3y ago • Edited 3y ago

I'm one of these guys that have built computers on schematics many many years ago... In '80ies. And I'm someone playing with idea to make one of RC2014... I think it is z80kits.com

Turing6502 • 3y ago

I've done a series on building up the ZX81 starting with a Z80 + EPROM which may be of interest
https://www.youtube.com/playlist?list=PLjQDRjQfW-84WG47-5UjPz1BrXxc1acvd

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